

# assignment\_02

Bill Perry

## Ecological Statistics Assignment 02

In this assignment you will analyze a dataframe of various limnological parameters from 594 lakes in Iowa, Wisconsin and Michigan. This dataframe is a trimmed version of the full dataframe used by Cheruvelil et al. 2013 in a study of the interactions between landscape parameters and limnological variables. The full dataframe is available online: [Dryad Full Dataframe](#)

### There are 16 columns in the dataframe:

1. lake - Lake Name
2. longitude\_utm - longitude in utm format
3. latitude\_utm - latitude in utm format
4. state - state abbreviation
5. source - source of data
6. state\_lake\_id - lake ID for the state
7. epa\_lake\_id - lake Id for EPA
8. date - data sample taken
9. year - year
10. CHL\_ugL - chlorophyll a concentration ( $\mu\text{g/L}$ )
11. TN\_ugL - total nitrogen (TN) concentration ( $\mu\text{g/L}$ )
12. TP\_ugL - total phosphorus (TP) concentration ( $\mu\text{g/L}$ )
13. Area\_km2 - lake area ( $\text{km}^2$ )
14. Z\_mean\_m - lake mean depth (m)
15. Waterhshed\_km2 - watershed area ( $\text{km}^2$ )
16. AgUrb.prop - categorical variable describing portion of watershed that is agricultural or urban (low; 0-49.99%, high; 50-100%).



Tip

**Note: When doing a multiple regression and the assumptions are not met, often the best approach is to take the log of the response variable - chl\_ugl in this case - and see how the model improves. The next approach is to do a log of each variable and see how the AIC values improve or take the log of all values. For the sake of this class you can do the log of all variables and see how it improves the AIC values (in a separate script I found that the log of TN, TP, and area were the optimal variables to transform. I strongly suggest doing a log of the chlorophyll a and leaving the rest along for ease of interpretation. You will also need to conver AgUrb.prop to a 0 and 1 variable**

One of the limnological parameters lake ecologists care most about is planktonic chlorophyll *a* concentration. Chlorophyll *a* concentration in the water is an indicator of the biomass of phytoplankton, the tiny algae that form

an important part of the base of aquatic food webs. Some algal growth is a good thing; too much algal growth can be associated with water quality problems like cyanobacterial blooms (which can produce toxins), oxygen depletion, fish kills, and reduced aesthetic value of lakes (think green, scummy, gross lakes). Lakes with too much algal growth are known as “eutrophic” lakes. An important question limnologists and aquatic ecologists have been grappling with in the past century is understanding the causes of eutrophication (excessive algal growth).

### **Your primary task is to answer the following questions:**

Your task is to use the trimmed Cheruvelil et al. (2013) dataset to examine patterns in chlorophyll a concentration in the different lakes and to use multiple linear regression and model comparisons to come up with the best model for predicting chlorophyll a concentration in a lake based on limnological (TP, TN, lake area, lake mean depth) and watershed (watershed area, land use) characteristics. Thus, chlorophyll concentration is the dependent (response) variable and other parameters are potential predictor (independent) variables.

### **To complete the task you will submit two files**

1. one containing your write-up
2. another containing fully functional and annotated R code that was used to generate summary statistics, figures and perform analyses.

Your write-up will include the following sections:

### **Your write-up will include the following sections:**

#### **Hypothesis statements**

- In this section you will state the hypotheses you are testing in this assignment in verbal form as well as in mathematical symbol notation.

#### **Statistical methods description**

- This section will include a detailed description of the statistical analyses you performed:
  - assumptions of tests
  - how these assumptions were checked
  - name and version of the software used to perform analyses (R and the libraries used)

#### **Statistical results description**

In this section you should present a description of the data, report the main trends and results of statistical analyses (together with information about results of assumption testing) in text form (use standard format for reporting results of tests) and refer the reader to any figures and tables summarizing the data. Each figure or table presented **must** be explicitly mentioned in this section, with a short description of the information contained in the table or figure. Remember to include statements on what your results mean for the hypotheses you are testing.

#### **Tables and figures**

Here you will place all the figures and tables mentioned in the text. Tables go first, followed by figures. Tables and figures should be placed in the same order in which they are mentioned in the body of the text. Each table or figure should be accompanied by an informative caption that will allow the reader to understand what is shown in the table/figure without reference to the text. Captions go below figures and above tables.

#### **Appendix**

This section will contain any additional figures (e.g., residual and qq plots). All these figures should also be referred to in the text in the appropriate order.

### **Assignment Tasks:**

- In the methods section, justify any pre-analysis decisions to drop variables as redundant, add variables, combine variables, or include interactions.

- A scatterplot matrix with all the variables that you initially decided to include in the model (before multicollinearity assessment). Add loess smoothers to the data in the lower left portion of the plot and correlation coefficients (and associated p-values) in the upper right. Justify the decision to use a particular correlation coefficient (Pearson, Spearman, Kendall).
- A table with VIF values for all the variables that you intended to include in the original model.
- If you decide to drop/modify/combine any variables based on results of multicollinearity assessment provide the correlation matrix and VIFs for the “final” set of variables that you will allow into the model.
- Partial regression plots (from the `car` package) that show the relationship between a predictor variables and chlorophyll a while holding the other predictors constant. Make sure the figure is properly labelled (see `?avPlots` for details on how to change the look of graphics).
- If you decide that the partial regression plots indicate non-linearity, describe the transformations that were used to approximate linearity for the predictor variables and provide a second set of partial regression plots showing that the non-linearity issue was addressed.
- Provide evidence that your response variable values, as specified in your first full model, meet (or don’t meet) the assumptions of multiple linear regression. If a transformation was necessary, explain what transformation was used and provide evidence it helped address assumption violations issues.
- Once a full model has been specified you will need to go through a model simplification process, dropping the non-significant term with the highest p-value (lowest F-value) and rechecking the results of the output until all remaining predictor parameters are significant. Provide a table summarizing all the models that were tested, starting with the full model and ending with the final, best model. Include full model F-ratios, p-values adjusted  $r^2$  and AIC values to compare the models.
- Model Number
- Model terms
- F-ratio
- p-values
- Adjusted  $r^2$
- AIC
- Remember!!! The `summary(model)` command will give you information about the significance of different levels of a categorical variable. The `anova(model)` command will tell you if the categorical predictor variable is significant overall.
- Provide a table with the dfs, F-ratios, p-values, and partial  $r^2$  values for the terms remaining in your final, “best” model.
- Use any other tables or figures that may help readers understand patterns in the data and the results of your analyses.
- Remember to include a statement in your results section that verbally summarizes the overall findings of your analysis in a form that will be understandable without reference to tables or figures.

## Grading Rubric:

Total is 100 points

- Hypotheses Statements - 10 points
- Statistical Methods - 20 points
- Results Statement - 30 points
- Figure and Table Quality - 20 points

- R Code - 16 points
- Grammar/Writing - 4 points

**References:**

Cheruvilil, K.S., P.A. Soranno, M.T. Bremigan, and K.E. Webster. 2013. Multi-scaled drivers of ecosystem state: Quantifying the importance of the regional spatial scale. *Ecological Applications* 23: 1603-1618.